

- 7 (a) State what is meant by a fundamental particle.

.....
 [1]

- (b) A nucleus X has 14 nucleons and p protons. The ratio of charge to mass for nucleus X is $4.1 \times 10^7 \text{ C kg}^{-1}$.

- (i) Determine p .

$p =$ [3]

- (ii) Nucleus X undergoes β^- decay to form nucleus Z.

Complete the equation representing this decay.



[3]

- (c) A sample of a radioactive substance emits particles that are positively charged and have a continuous range of kinetic energies.

State and explain whether the nuclei in the sample are undergoing α -decay, β^+ decay or β^- decay.

.....

 [2]

[Total: 9]

